CS 401 Artificial Intelligence

Expert Systems

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Expert Systems

- An expert system is a computer program that contains knowledge and solves problems in a specific field in much the same way that a human expert would.

- It stores the knowledge of one or more human experts in a particular field.

Expert Systems

- The field is called a domain. The experts are called domain experts. A user presents the expert system with the specifics of a problem within the domain.

- The system applies its stored knowledge to solve the problem.

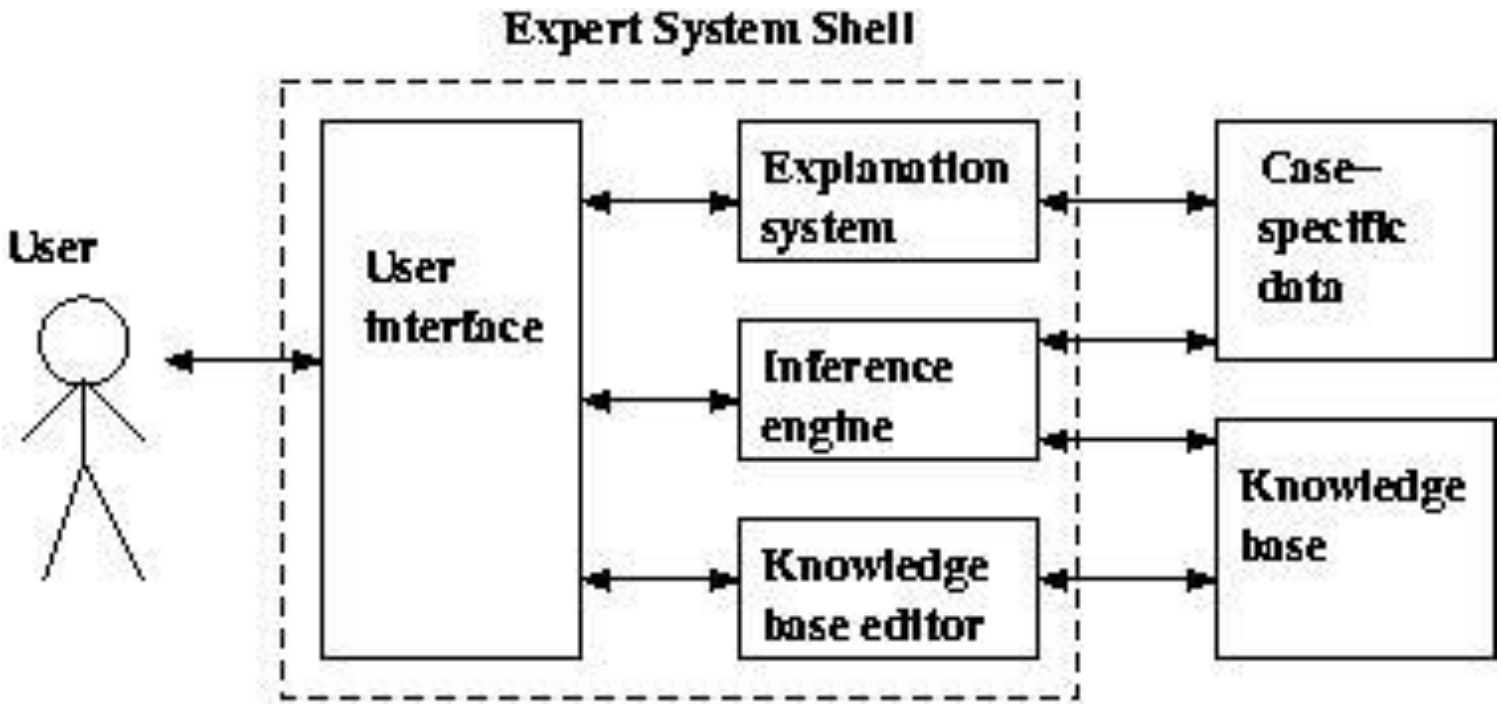
Expert Systems

- Human experts are able to perform at high level because they know a lot about their area of expertise.
- This simple observation is the underlying rationale for the design of *strong method* or *knowledge-based* problem solvers.
- An expert system, for example, uses knowledge specific to a problem domain to provide “expert quality” performance in that application area.
- Generally expert system designers acquire this knowledge with the help of human domain experts, and the system emulates the human expert’s methodology and performance.

Expert systems are built to solve a wide range of problems in domains such as medicine, mathematics, engineering, chemistry, geology, computer science, business, law, defense and education. These programs address a variety of problems, the following list, from Waterman, is a useful summary of general expert system problem categories.

- ***Interpretation*** – forming high-level conclusions from collection of raw data.
- ***Prediction*** – projecting probable consequences of given situation.
- ***Diagnosis*** – determine the cause of malfunctioning in complex situations based on observable symptoms.
- ***Design*** – finding a configuration of system components that meet performance goals while satisfying a set of design constraints.
- ***Planning*** – devising a sequence of actions that will achieve a set of goals given certain starting conditions and run-time constraints.
- ***Monitoring*** – comparing a system's observed behavior to its expected behavior.
- ***Instruction*** – assisting in the education process in technical domains.
- ***Control*** – governing the behavior of a complex environment.

Architecture of an Expert System



Components of an Expert System

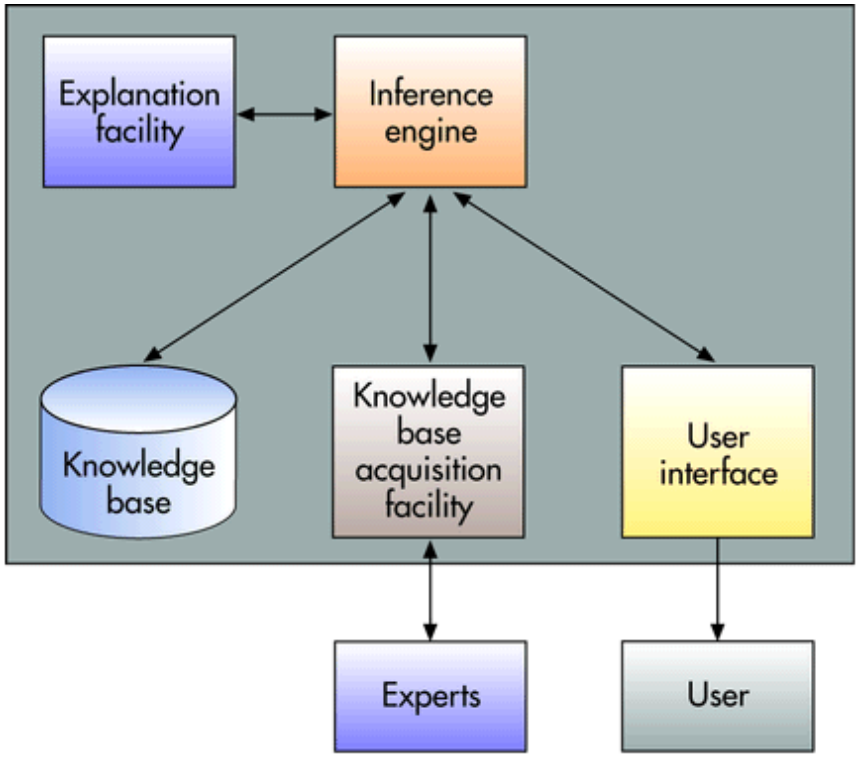


FIGURE 11.7

Components of an Expert System

Components of an Expert System

- For a rule-based expert system: The user interacts with the system through a *user interface* which may use menus, natural language or any other style of interaction.
- Then an *inference engine* is used to reason with both the *expert knowledge* (extracted from our expert) and data specific to the particular problem being solved. The expert knowledge will typically be in the form of a set of IF-THEN rules.
- The *case specific data* includes both data provided by the user and partial conclusions (along with certainty measures) based on this data. In a simple forward chaining rule-based system the case specific data will be the elements in *working memory*.

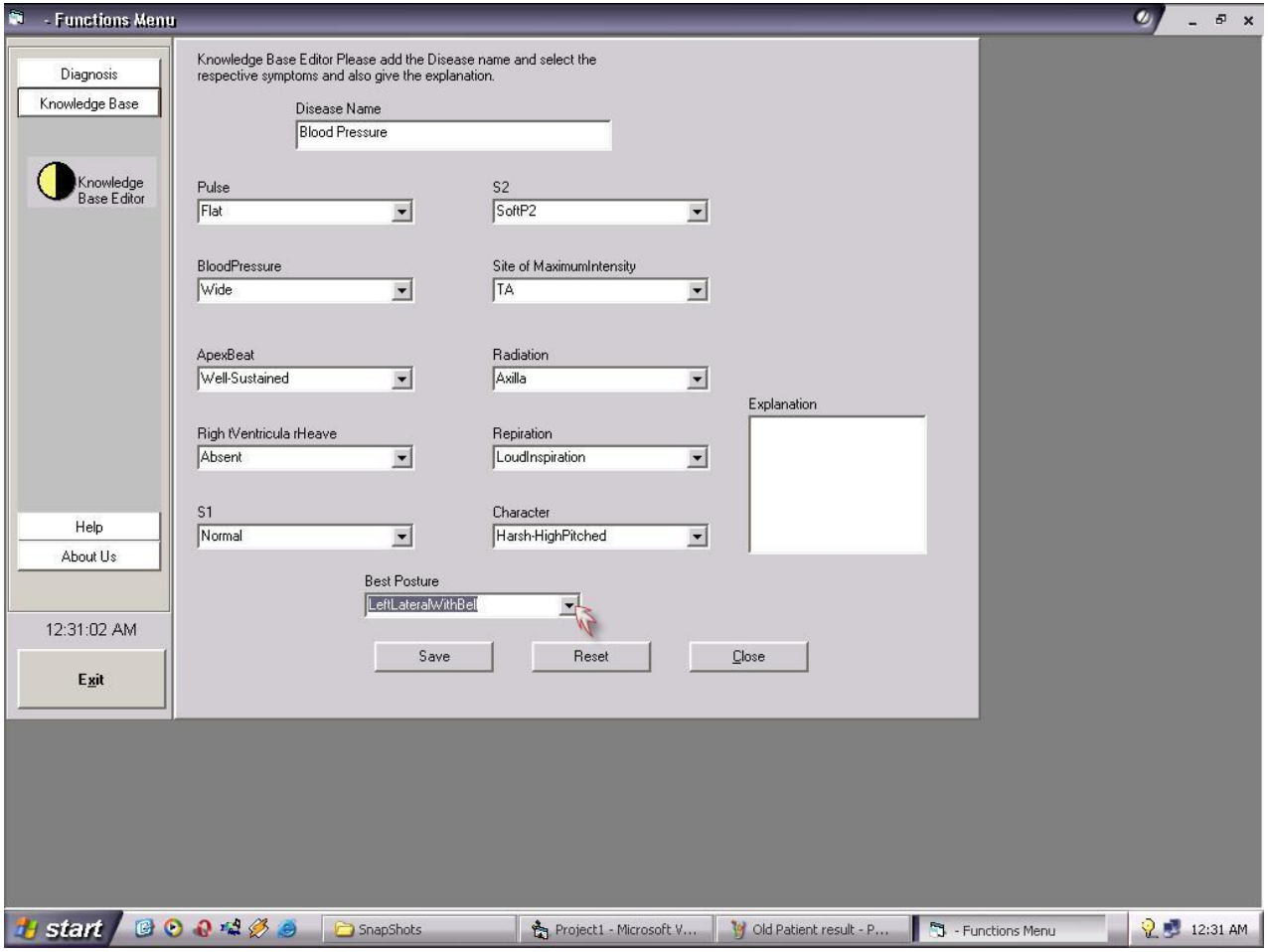
Components of an Expert System

- Almost all expert systems also have an *explanation subsystem*, which allows the program to explain its reasoning to the user. Some systems also have a *knowledge base editor* which helps the expert or knowledge engineer to easily update and check the knowledge base.

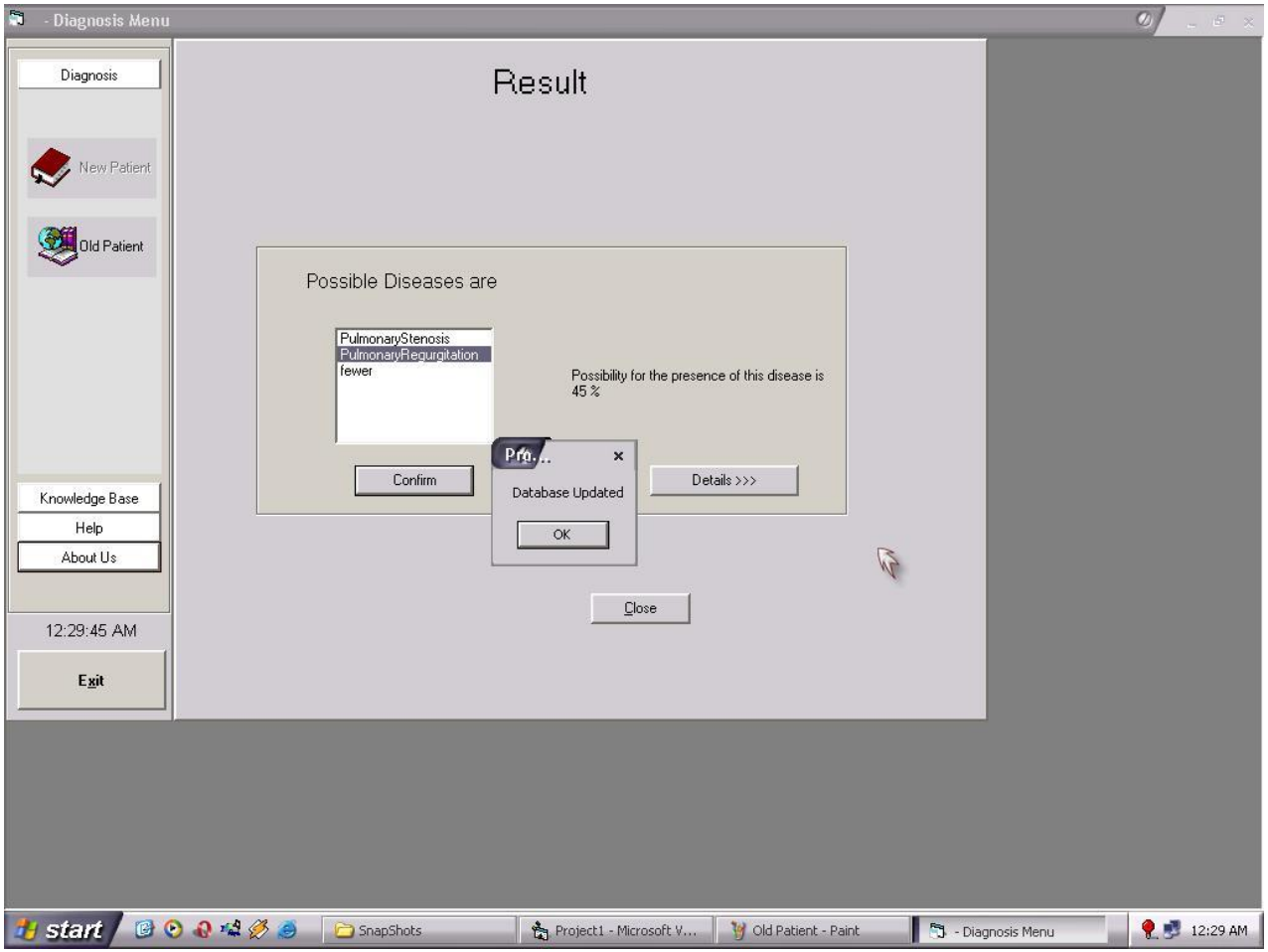
Expert System Shell

- One important feature of expert systems is the way they (usually) separate domain specific knowledge from more general purpose reasoning and representation techniques. The general purpose bit (in the dotted box in the figure) is referred to as an *expert system shell*.
- As we see in the figure, the shell will provide the inference engine (and knowledge representation scheme), a user interface, an explanation system and sometimes a knowledge base editor.
- Given a new kind of problem to solve (say, car design), we can usually find a shell that provides the right sort of support for that problem, so all we need to do is provide the expert knowledge.
- There are numerous commercial expert system shells, each one appropriate for a slightly different range of problems.
- Using shells to write expert systems generally greatly reduces the cost and time of development (compared with writing the expert system from scratch).

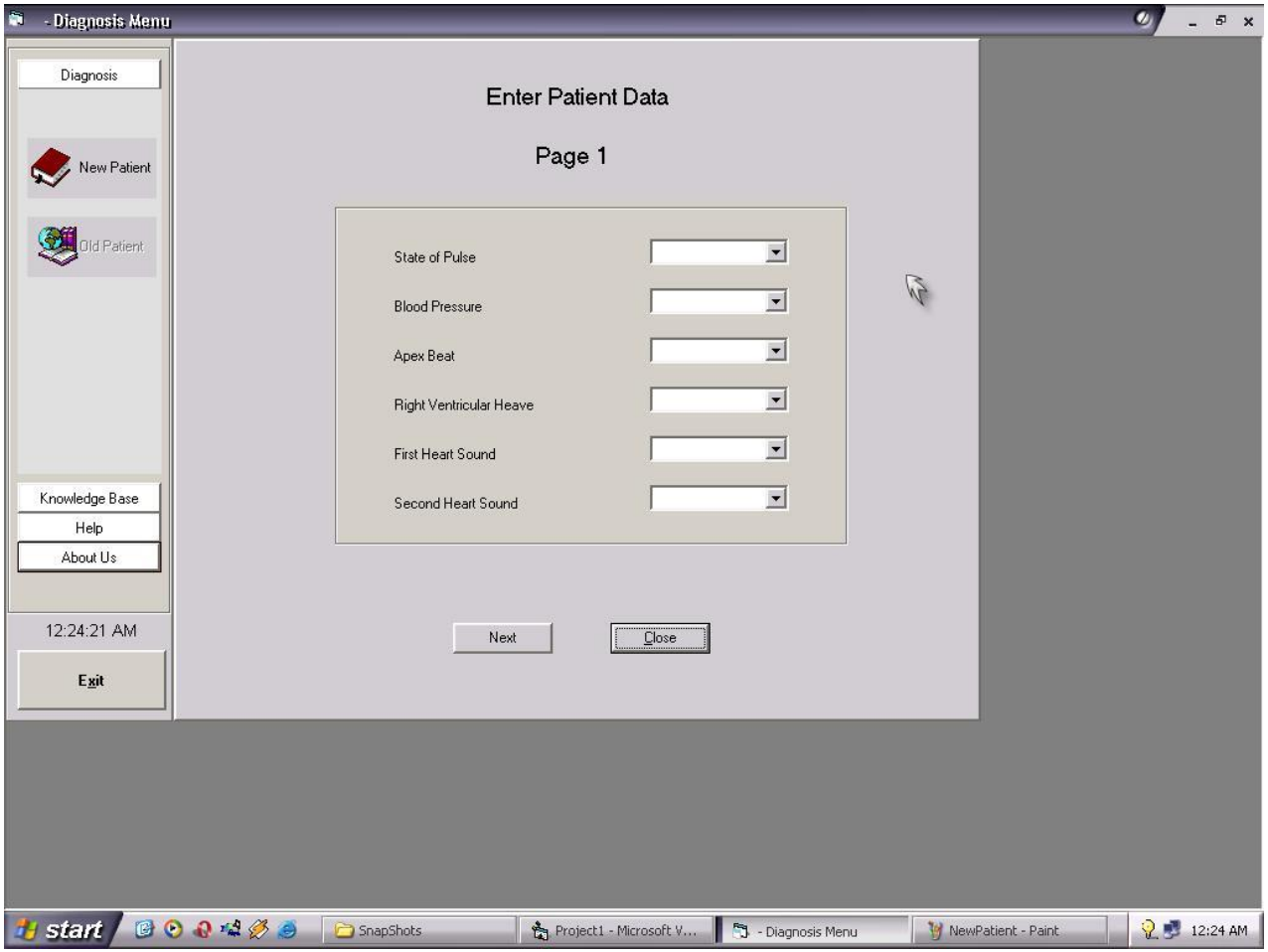
Example: Medical Expert System



Example: Medical Expert System



Example: Medical Expert System



Rule Based Reasoning Systems

Inference Engine:

Knowledge processor modelled after the expert's reasoning. It matches the facts contained in the working memory with the domain knowledge contained in the knowledge base, to draw conclusions about the problem

Example: recognize-act cycle of Production Systems

Rule Based Reasoning Systems

Working Memory:

Part of an expert system that contains the problem facts discovered during the session.
It is a model of short term memory of human experts.
Contains all information supplied by the user or inferred by the system

Knowledge Base

IF the ball's colour is red
THEN I like the ball

IF I like the ball
THEN I will buy the ball

Working Memory

Ball's colour is red

I like the ball

I will buy the ball

Rule Based Reasoning Systems

Explanation Facility:

This module provides an explanation to the user about "why" an expert system is asking a question and "how" it reached a conclusion

Rule Based Reasoning Systems

Why Build an Expert System:

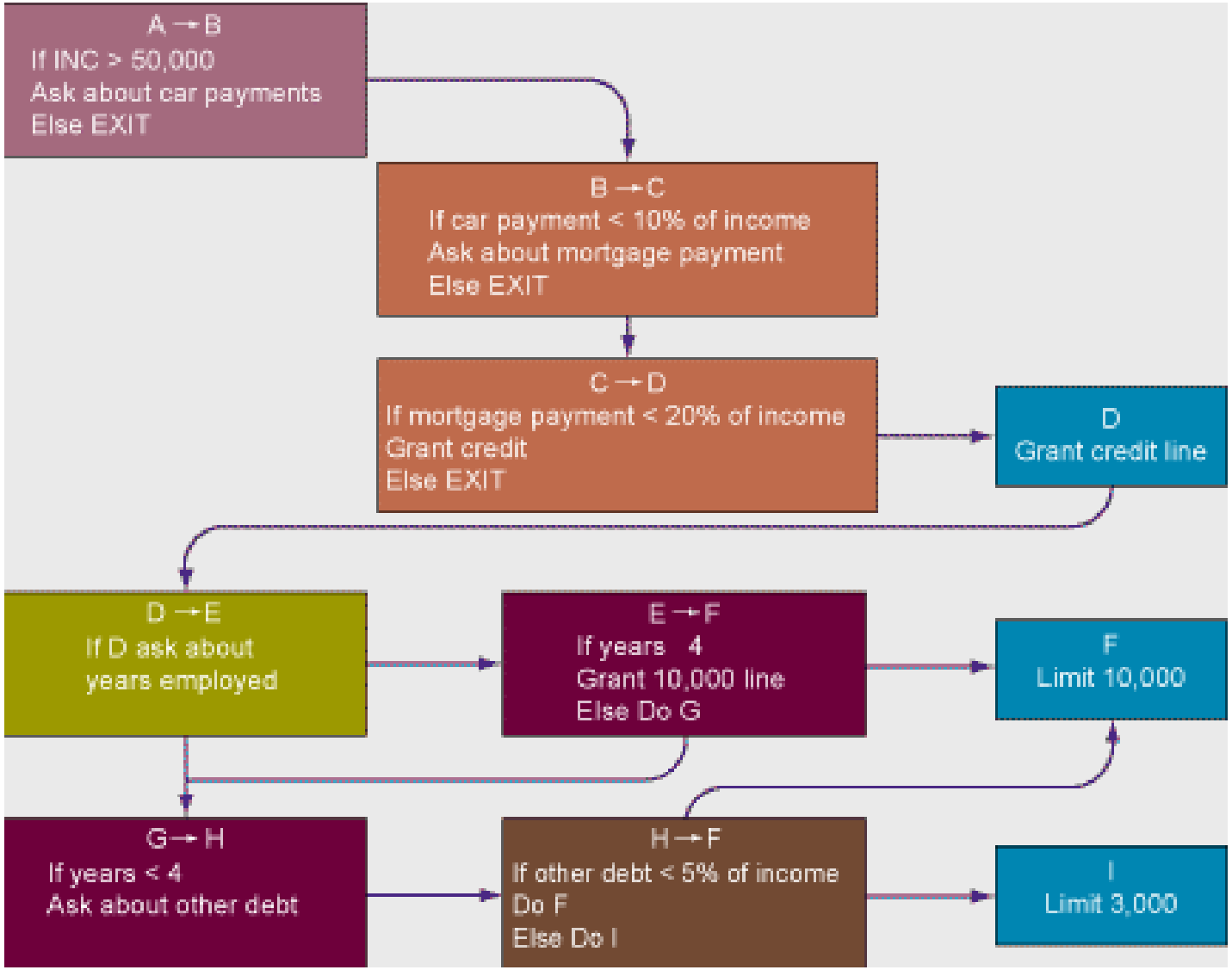
Factor	Human	Expert System
Time availability	Workday	Always
Geographic	Local	Anywhere (soft copies)
Safety (hostile environments)	Irreplaceable	Replaceable
Perishable (death, retirement, job transfer)	Yes	No

Rule Based Reasoning Systems

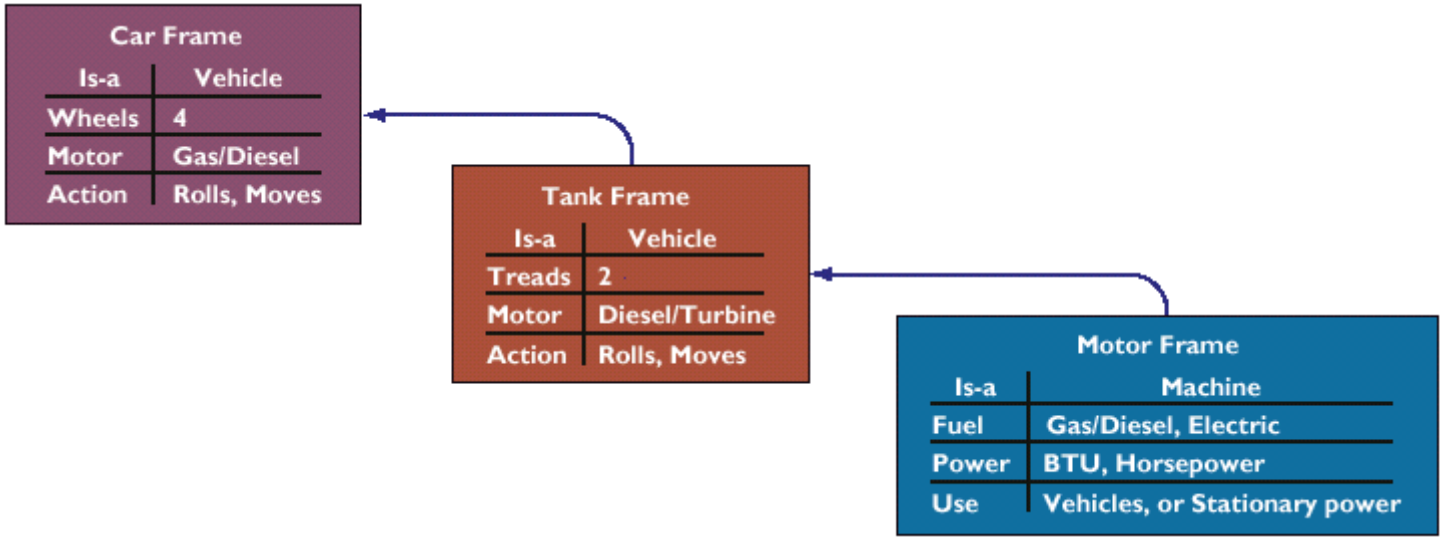
Why Build an Expert System:

Factor	Human	Expert System
Performance	Variable	Consistent
Speed	Variable	Consistent (usually faster)
Cost	High	Affordable (development costs high, but recovered quickly)

Rules in an AI program

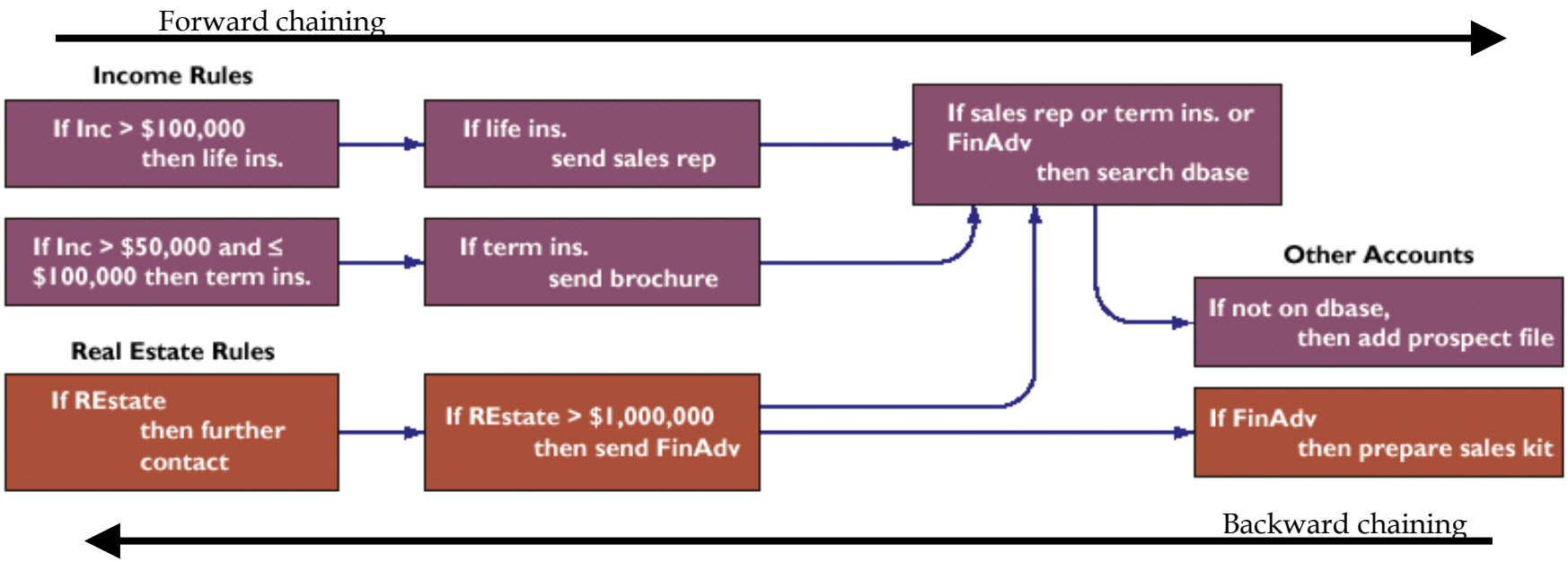


Frames to Model Knowledge



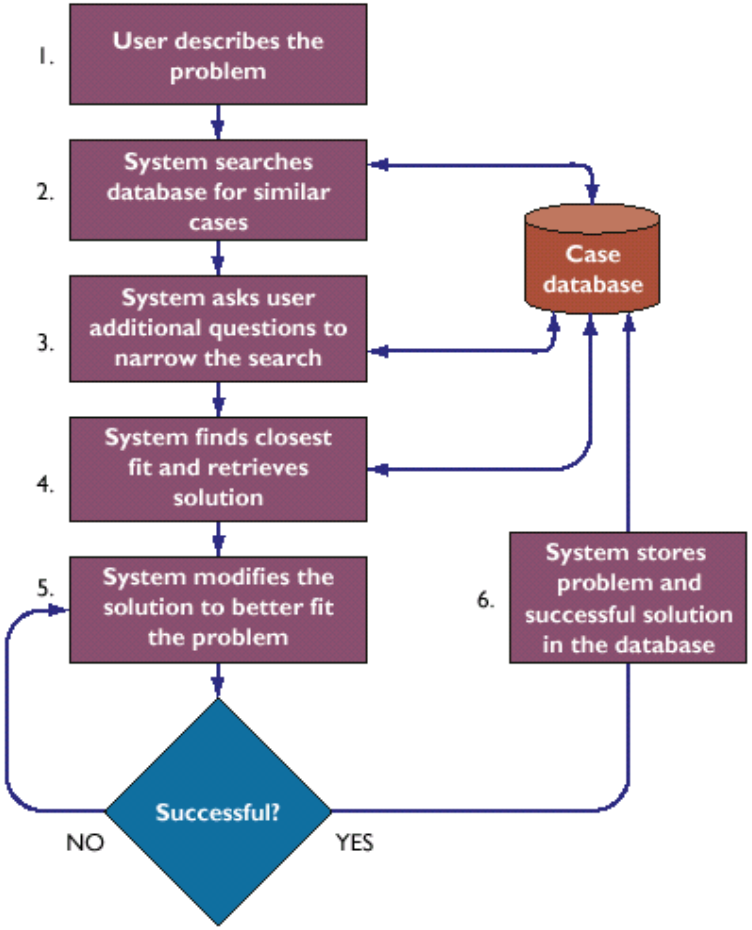
Capturing Knowledge: Expert Systems

- Inference Engine (Chaining)



Capturing Organizational Knowledge

- Case-based Reasoning (CBR)



Reading Assignment & References

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Section 7.0, - 7.3.4 Page 247-283
Few examples:

http://www.exsys.com/demomain.html

http://www.aistudy.co.kr/demo/es_demo.htm

http://www.kbsc.com/rulebase.html

http://www.expertise2go.com/webesie/

https://en.wikipedia.org/wiki/Expert_system